

**Master of Computer Application (M.C.A) (Two Year Programme) - 2nd
Sem. (Credit Based Evaluation and Grading System) (Batch 2024-26)
(2525)**

Paper: CSL-4033 Microprocessor & Its Applications

Time allowed: 3 hrs.

Max. Marks: 70

Note: Attempt five questions selecting at least one question from each Section. The fifth question may be attempted from any Section.

Section-A

- Q1 a) Differentiate the role and features of Microcontroller and Micro-processors. 7
 b) Discuss the integration of Bus interface Unit (BIU) and execution unit (EU) of a micro-processor system. 7
- Q2 a) How Segment registers and index registers are used during program execution by a micro-processor? Explain. 8
 b) Discuss the role of Carry, sign and parity flag and bits of status register. 6

Section-B

- Q3 How maximum mode of 8086 microprocessor can be configured? Explain the pins used exclusively by maximum mode signals. 14
- Q4 a) How address space is used? Why memory is divided into two banks in case of 8086? Explain. 7
 b) Discuss the role of write memory bus cycle used by 8086 configured as minimum mode. 7

Section-C

- Q5 Discuss the role of Programmable Peripheral interface (PPI). How this interface is used for I/O communication? Explain. 14
- Q6 Discuss the mechanism of IVT used by interrupts. Explain the role of different types of interrupts of 8086. 14

Section-D

- Q7 a) Discuss the general structure for development of assembly language program. 14
 b) What are the addressing modes? How are they used by instruction set? Explain. 14
- Q8 Explain the following:
 a) Data movement instructions 7
 b) Logical instructions 7

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